

Mini Project Report

Description: Structure Documentation for Mini project

Project Title

End-to-End Machine Learning Application with Streamlit Deployment

Problem Statement

Many machine learning projects fail to demonstrate complete real-world workflow, often focusing only on model training. This project aims to design and implement a complete ML pipeline including data preprocessing, model training, evaluation, and deployment through a user-friendly web interface.

Project Description

This project implements an end-to-end Machine Learning application using Python. It covers data cleaning, feature preparation, model training, testing, hyperparameter tuning, and deployment using Streamlit. The application allows users to interact with the ML model through a simple UI.

Objectives

- Preprocess and clean raw datasets
- Build and test machine learning models
- Evaluate model performance using train-test split
- Optimize models using hyperparameter tuning
- Deploy the model using Streamlit UI

System Workflow

1. Data Collection
2. Data Cleaning and Preprocessing
3. Feature Engineering
4. Model Training
5. Model Evaluation
6. UI Integration using Streamlit
7. Result Visualization

Tools and Technologies

- Python – Core programming language
- Pandas, NumPy – Data preprocessing
- Scikit-learn – ML modeling
- CatBoost, LightGBM – Advanced models
- Optuna – Hyperparameter tuning
- Streamlit – Web-based deployment

Implementation Timeline

Week 1: UI design, dataset cleaning, ML pipeline structure

Week 2: Model training, testing, evaluation, and deployment

Results

The system successfully processes datasets, trains ML models, and generates predictions through a Streamlit-based interface.

Future Enhancements

- Deploy application
- Add real-time data input
- Improve UI design
- Implement model monitoring

Conclusion

This project demonstrates practical understanding of Machine Learning workflows and deployment. It strengthens skills in data handling, ML modeling, and UI integration.